# The waist class is quadratic, not additive, under band composition on $T^2$

MacBeth
Kodamai

2026-10-14

**Author:** MacBeth (Kodamai). **Date:** 2026-10-14.
**Prepared for** Rick (referee) and Neil Ghani (PI); **CC** Robin Langer.
**Content commit:** `0edf2e6`.
**Provenance.** Registry node `delta-additive-in-band` [**proved**] in registry `supply-chain-inventory-gerbe-h2`; child of `coupled-T2-general-delta-lemma` [**proved**] and `gerbe-two-ends-correlation` [**proved**]. Proof file `proofs/2026-10-13-delta-additive-in-band.md`; verification script `scratch/2026-10-13-delta-additivity.py` (couplers $D_4 \times C_2$, $Q_8 \times C_2$; 0 violations on every clause below). This note typesets a complete, registry-graded result; it introduces no new claim beyond the cited proof file.

**Abstract**

Fix base $C = B(\mathbb{Z}^2)$ (nerve $T^2$) and a structure group $K$ of nilpotency class $\leq 2$, so that $M := \mathrm{Inn}K$ is abelian. The general $\delta$-commutator lemma assigns to a band $\rho\colon \mathbb{Z}^2 \to \mathrm{Out}K$ — a commuting pair of outer classes — the connecting class $\delta(\rho) = [[A, B]] \in H^2(T^2; M)$ for any lifts $A, B \in \mathrm{Aut}K$. We determine how $\delta$ behaves under band composition. The answer is: *not additively*. A master cross-term formula (Theorem 1), an on-the-nose identity in $\mathrm{Inn}K$, computes the composite commutator in full. On the trivial-action locus $\mathrm{Out}^0$, where the additivity question is well-typed, this collapses to a quadratic law $\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2) + \chi$ with symmetric bi-additive polar form $\chi$ (Theorem 5), whence $\delta(\rho^n) = n^2\delta(\rho)$. A dichotomy (Theorem 6) shows $\delta$ is additive on an abelian subgroup iff it is identically zero there: *there is no abelian subgroup on which $\delta$ is a nonzero homomorphism*, so a coupled torus is always a witness to non-additivity. Coupling, non-splitting of the lift extension, and non-additivity of the waist class are one and the same phenomenon. All clauses are verified computationally with 0 violations on the two $T^2$ couplers.

## 1 The question and the answer

Fix base $C = B(\mathbb{Z}^2)$ (nerve $T^2$) and a structure group $K$ of nilpotency class $\leq 2$, so $M := \mathrm{Inn}K$ is abelian. By the general $\delta$-commutator lemma (`coupled-T2-general-delta-lemma`, proved 2026-10-12), a band $\rho\colon \mathbb{Z}^2 \to \mathrm{Out}K$ — i.e. a commuting pair $\bar{A} := \rho(s)$, $\bar{B} := \rho(t) \in \mathrm{Out}K$ — has connecting class

$$\delta(\rho) = [\,[A, B]\,] \in H^2(T^2; M) = M_{\mathbb{Z}^2} = M\big/\big((\sigma - 1)M + (\tau - 1)M\big), \tag{1}$$

where $A, B \in \mathrm{Aut}K$ are *any* lifts of $\bar{A}, \bar{B}$ and $\sigma = \bar{A} \cdot (-)$, $\tau = \bar{B} \cdot (-)$ are the induced actions of $\bar{A}, \bar{B}$ on $M$ by conjugation.

Let $\rho_1, \rho_2$ be two bands landing in a *common abelian subgroup* $\bar{\mathcal{A}} \leq \mathrm{Out}K$ (the concrete meaning of "a common system of commuting lifts": their pointwise product $\rho_1\rho_2 = (\bar{A}_1\bar{A}_2,\ \bar{B}_1\bar{B}_2)$ is again a band into $\bar{\mathcal{A}}$, because $\bar{\mathcal{A}}$ is abelian). **Is $\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2)$?**

**Answer: (B) — No. $\delta$ is not a homomorphism.** The exact defect is governed by the master cross-term formula (Theorem 1), an on-the-nose identity in $\mathrm{Inn}K$:

$$[A_1A_2,\, B_1B_2] \;=\; \underbrace{[A_1, B_1]}_{=\delta(\rho_1)\text{ rep}} \;+\; (\sigma_1\tau_1)\cdot[A_2, B_2] \;+\; \sigma_1\cdot[A_2, B_1] \;+\; \tau_1\cdot[A_1, B_2].$$

On the *trivial-action locus* $\mathrm{Out}^0 = \ker(\mathrm{Out}K \to \mathrm{Aut}M)$ — where the question is even well-typed, all three classes living in the *same* group $M$ — this collapses (Theorem 5) to

$$\boxed{\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2) + \chi(\rho_1, \rho_2), \qquad \chi(\rho_1, \rho_2) = [A_1, B_2] + [A_2, B_1] \in M}$$

with $\chi$ a *symmetric bi-additive* form — the polarization of the quadratic map $\delta$, which obeys $\delta(\rho^n) = n^2\,\delta(\rho)$. And then the *dichotomy* (Theorem 6): on any abelian $\bar{\mathcal{A}} \subseteq \mathrm{Out}^0$,

$$\delta \text{ is additive on bands into } \bar{\mathcal{A}} \iff \delta \equiv 0 \text{ there} \iff \pi^{-1}(\bar{\mathcal{A}}) \leq \mathrm{Aut}K \text{ is abelian.}$$

**There is no abelian subgroup on which $\delta$ is a nonzero homomorphism.** A coupled torus is *always* a witness to non-additivity.

So the butterfly §7 open edge "composite weld obstruction = composite of waist classes?" resolves in the negative, in the sharpest possible form: the waist class $\delta$ is a *quadratic refinement*, not a homomorphism, on the composition monoid of bands; its polar form $\chi$ is the obstruction to additivity, and $\chi$ vanishes exactly when $\delta$ itself does.

## 2 The master cross-term formula

Throughout, $M = \mathrm{Inn}K \trianglelefteq \mathrm{Aut}K$ is abelian (class $\leq 2$), written additively, and for $X \in \mathrm{Aut}K$, $m \in M$ we write $X \cdot m := XmX^{-1} \in M$ ($M$ normal); this factors through $\pi(X) \in \mathrm{Out}K$. For a band $\rho_i$ fix lifts $A_i, B_i \in \mathrm{Aut}K$ of $\rho_i(s), \rho_i(t)$ and set $\sigma_i := \rho_i(s)\cdot(-)$, $\tau_i := \rho_i(t)\cdot(-) \in \mathrm{Aut}(M)$.

**Theorem 1** (master cross-term formula)**.** *Let $K$ be finite of class $\leq 2$ and let $\rho_1, \rho_2$ be bands with all four outer classes $\bar{A}_1, \bar{B}_1, \bar{A}_2, \bar{B}_2$ in a common abelian subgroup of $\mathrm{Out}K$. Then every pairwise commutator $[A_i, B_j]$ lies in $M$, and, additively in $M$,*

$$[A_1A_2,\, B_1B_2] = [A_1, B_1] + (\sigma_1\tau_1)\cdot[A_2, B_2] + \sigma_1\cdot[A_2, B_1] + \tau_1\cdot[A_1, B_2]. \qquad (\star)$$

*Since $A_1A_2$, $B_1B_2$ are lifts of $(\rho_1\rho_2)(s), (\rho_1\rho_2)(t)$, the left side represents $\delta(\rho_1\rho_2) \in M_{\mathbb{Z}^2}$ (coinvariants for the product action $\sigma_1\sigma_2, \tau_1\tau_2$).*

*Proof.* Membership: $\pi[A_i, B_j] = [\bar{A}_i, \bar{B}_j] = 1$ because $\bar{A}_i, \bar{B}_j$ commute (common abelian subgroup), so $[A_i, B_j] \in \ker\pi = M$. Use the universal commutator identities

$$[xy, z] = x[y, z]x^{-1}\cdot[x, z], \qquad [x, yz] = [x, y]\cdot y[x, z]y^{-1}.$$

Apply the first with $x = A_1, y = A_2, z = B_1B_2$:

$$[A_1A_2, B_1B_2] = A_1[A_2, B_1B_2]A_1^{-1}\cdot[A_1, B_1B_2].$$

Apply the second to each factor:

$$[A_2, B_1B_2] = [A_2, B_1] \cdot B_1[A_2, B_2]B_1^{-1}, \qquad [A_1, B_1B_2] = [A_1, B_1] \cdot B_1[A_1, B_2]B_1^{-1}.$$

Substituting and conjugating the first bracket by $A_1$:

$$[A_1A_2, B_1B_2] = \big(A_1[A_2, B_1]A_1^{-1}\big) \cdot \big(A_1B_1[A_2, B_2]B_1^{-1}A_1^{-1}\big) \cdot [A_1, B_1] \cdot \big(B_1[A_1, B_2]B_1^{-1}\big).$$

All four factors lie in the abelian group $M$, so we may reorder and pass to additive notation. The conjugating prefixes are $A_1$ (giving $\sigma_1$), $A_1B_1$ (giving $\sigma_1\tau_1$), nothing (giving 1), and $B_1$ (giving $\tau_1$). This is exactly $(\star)$. □

**Verification.** Script `scratch/2026-10-13-delta-additivity.py`, clause `[GEN]`: $(\star)$ holds *on the nose in* Inn$K$ over *all* quadruples of bands inside a common abelian subgroup — 8113 checks for $D_4 \times C_2$ and 26037 for $Q_8 \times C_2$, 0 **violations**. (An earlier run comparing against the representative lift of the *product coset* showed spurious "violations"; those are precisely coboundaries $(\sigma-1)M + (\tau-1)M$ — the product-coset representative differs from $A_1A_2$ by an inner automorphism — and vanish in $M_{\mathbb{Z}^2}$. Using the on-the-nose product lift $A_1A_2$ removes them, confirming $(\star)$ is an identity in $M$, not merely in the quotient.)

**Reading of $(\star)$.** Beyond the two "diagonal" terms $\delta(\rho_1), \delta(\rho_2)$ there are: a *twist* $(\sigma_1\tau_1)$ acting on $\delta(\rho_2)$'s representative, and the two *cross terms* $\sigma_1[A_2, B_1] + \tau_1[A_1, B_2]$. Both the twist and the asymmetry ($\sigma_1, \tau_1$ but not $\sigma_2, \tau_2$) are artifacts of writing the product lift as $A_1A_2$ rather than $A_2A_1$; they disappear precisely when the action is trivial.

## 3 The trivial-action locus: $\delta$ is a quadratic map

**Definition 2.** $\mathrm{Out}^0 := \ker\big(\mathrm{Out}K \to \mathrm{Aut}(M)\big)$ is the subgroup of outer classes acting trivially on $M = \mathrm{Inn}K$. A band $\rho$ is *trivial-action* iff $\bar{A}, \bar{B} \in \mathrm{Out}^0$, i.e. $\sigma = \tau = \mathrm{id}$.

**Remark 3** (terminology)**.** The PROVE brief's "trivial centre-action" is here made precise as the action relevant to $\delta$: trivial action of Out$K$ on the coefficient module $M = \mathrm{Inn}K$ — the exact condition under which $\delta$'s receptacle $M_{\mathbb{Z}^2}$ collapses to $M$ and the additivity question becomes well-typed. For class-$\leq 2$ $K$ this is also the condition making the restricted extension $(*)|_{\mathrm{Out}^0}$ central, below. We do *not* silently conflate it with the distinct condition "Out$K$ acts trivially on $Z(K)$"; the relevant module for the waist class is Inn$K$, not $Z(K)$.

Over $\mathrm{Out}^0$ the restricted extension $1 \to M \to \mathrm{Aut}^0K \to \mathrm{Out}^0 \to 1$ (with $\mathrm{Aut}^0K := \pi^{-1}(\mathrm{Out}^0)$) is a *central* extension: $M$ is central in $\mathrm{Aut}^0K$ (verified: clause `restricted (*) central: True` for both couplers). For such a central extension the *commutator pairing*

$$c\colon \{\text{commuting pairs in } \mathrm{Out}^0\} \to M, \qquad c(\bar{x}, \bar{y}) := [\tilde{x}, \tilde{y}]$$

is well defined (lift-independent: for central $u, v \in M$, $[\tilde{x}u, \tilde{y}v] = [\tilde{x}, \tilde{y}]$) and $\delta(\rho) = c(\rho(s), \rho(t))$ for trivial-action bands, with $M_{\mathbb{Z}^2} = M$ (no coboundaries).

**Lemma 4** ($c$ is alternating and bi-multiplicative)**.** *On commuting pairs in* $\mathrm{Out}^0$*:* $c(\bar{x}, \bar{x}) = 0$*;* $c(\bar{y}, \bar{x}) = -c(\bar{x}, \bar{y})$*; and whenever all pairs in sight commute,*

$$c(\bar{x}\bar{y}, \bar{z}) = c(\bar{x}, \bar{z}) + c(\bar{y}, \bar{z}), \qquad c(\bar{x}, \bar{y}\bar{z}) = c(\bar{x}, \bar{y}) + c(\bar{x}, \bar{z}).$$

*Proof.* Alternating/antisymmetry are immediate from $[\tilde{x},\tilde{x}]=1$, $[\tilde{y},\tilde{x}]=[\tilde{x},\tilde{y}]^{-1}$. For bi-multiplicativity,

$$[\tilde{x}\tilde{y},\tilde{z}] = \tilde{x}[\tilde{y},\tilde{z}]\tilde{x}^{-1}\,[\tilde{x},\tilde{z}] = [\tilde{y},\tilde{z}][\tilde{x},\tilde{z}],$$

using that $[\tilde{y},\tilde{z}] \in M$ is *central* (this is exactly the trivial-action hypothesis). □

(Clause `(2) bi-mult on Out^0`: 0 violations; and `(2)` over *all* of Out **fails** — 112/1408 and 962/4032 violations — confirming centrality is essential.)

**Theorem 5** (quadratic law + cross term)**.** *For trivial-action bands $\rho_1,\rho_2$ into a common abelian $\bar{\mathcal{A}} \subseteq \mathrm{Out}^0$, equality holds in $M$:*

$$\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2) + \chi(\rho_1,\rho_2), \qquad \chi(\rho_1,\rho_2) = c(\bar{A}_1,\bar{B}_2) + c(\bar{A}_2,\bar{B}_1) = [A_1,B_2] + [A_2,B_1].$$

*$\chi$ is symmetric ($\chi(\rho_1,\rho_2) = \chi(\rho_2,\rho_1)$) and bi-additive; $\delta$ is a* quadratic map $\mathrm{Hom}(\mathbb{Z}^2,\bar{\mathcal{A}}) \cong \bar{\mathcal{A}}\times\bar{\mathcal{A}} \to M$ *with polar form $\chi$, and in particular $\delta(\rho^n) = n^2\,\delta(\rho)$ and $\delta(\rho^{-1}) = \delta(\rho)$.*

*Proof.* Either specialize ($\star$) to $\sigma_1 = \tau_1 = \mathrm{id}$, or compute directly with Lemma 4:

$$\delta(\rho_1\rho_2) = c(\bar{A}_1\bar{A}_2, \bar{B}_1\bar{B}_2) = c(\bar{A}_1,\bar{B}_1) + c(\bar{A}_1,\bar{B}_2) + c(\bar{A}_2,\bar{B}_1) + c(\bar{A}_2,\bar{B}_2) = \delta(\rho_1) + \delta(\rho_2) + \chi.$$

Symmetry of $\chi$ is visible; bi-additivity is bi-multiplicativity of $c$. Quadraticity: $\delta(\rho^n) = c(\bar{A}^n,\bar{B}^n) = n^2 c(\bar{A},\bar{B}) = n^2\delta(\rho)$ by bi-multiplicativity; $\delta(\rho^{-1}) = c(\bar{A}^{-1},\bar{B}^{-1}) = (-1)(-1)c(\bar{A},\bar{B}) = \delta(\rho)$. □

**Verification.** Clause `(3) cross-term on Out^0` (1905 checks each) and `(4) quadratic` (320 checks each, $n \le 6$): 0 **violations** on both couplers.

## 4 The additivity dichotomy

**Theorem 6** (dichotomy)**.** *Let $\bar{\mathcal{A}} \subseteq \mathrm{Out}^0$ be abelian. The following are equivalent:*

1. *$\delta$ is a group homomorphism on* $\mathrm{Hom}(\mathbb{Z}^2,\bar{\mathcal{A}})$ *(additivity);*
2. *$\delta \equiv 0$ on all bands into $\bar{\mathcal{A}}$;*
3. *$c|_{\bar{\mathcal{A}}\times\bar{\mathcal{A}}} \equiv 0$;*
4. *$\pi^{-1}(\bar{\mathcal{A}}) \le \mathrm{Aut}K$ is abelian (i.e. $\bar{\mathcal{A}}$ has a system of commuting lifts).*

*In particular $\delta$ is* never a nonzero homomorphism *on any abelian subgroup.*

*Proof.* (2) ⇒ (1) trivial. (1) ⇒ (3): additivity forces $\chi \equiv 0$; take $\rho_1 = (\bar{x},1), \rho_2 = (1,\bar{y})$, so $\delta(\rho_1) = c(\bar{x},1) = 0$, $\delta(\rho_2) = c(1,\bar{y}) = 0$, while $\rho_1\rho_2 = (\bar{x},\bar{y})$ gives $\delta(\rho_1\rho_2) = c(\bar{x},\bar{y})$; additivity makes this 0 for all $\bar{x},\bar{y} \in \bar{\mathcal{A}}$. (3) ⇒ (2): $\delta(\rho) = c(\bar{A},\bar{B}) = 0$. (3) ⇔ (4): $\pi^{-1}(\bar{\mathcal{A}})$ is a central extension of the abelian $\bar{\mathcal{A}}$ by $M$, hence abelian iff its commutator pairing $c|_{\bar{\mathcal{A}}}$ vanishes. □

**Verification.** Clause `(5) dichotomy on Out^0`: among the 6 coupled abelian subgroups of $\mathrm{Out}^0$ (for each coupler), the count `nonzero & ADDITIVE` is 0; every coupled subgroup is non-additive. Exhaustive over ≤ 2-generated abelian subgroups (which suffices, bands being rank ≤ 2).

**Reconciliation with the coupling criterion.** (4) is the negation of the $\delta$-lemma's coupling criterion "$\exists$ commuting outer pair with no commuting lift." So Theorem 6 says:

$$\bar{\mathcal{A}} \text{ couples (some band has } \delta \neq 0) \iff \pi^{-1}(\bar{\mathcal{A}}) \text{ nonabelian} \iff \delta \text{ is non-additive on } \bar{\mathcal{A}}.$$

Coupling, non-splitting of $\pi^{-1}(\bar{\mathcal{A}})$, and non-additivity of the waist class are *one and the same phenomenon*.

## 5 Scope, and the genuinely twisted regime

The clean additivity package (Theorems 5–6) lives on the *trivial-action locus* $\mathrm{Out}^0$, and this is not a cosmetic restriction but the precise place the question is *well-typed*: for bands with differing actions, $\delta(\rho_1), \delta(\rho_2), \delta(\rho_1\rho_2)$ lie in *different* coinvariant quotients $M_{(\sigma,\tau)}$, so "$\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2)$" is not even a statement about one group until a common receptacle is chosen.

**Coupling is *not* confined to $\mathrm{Out}^0$.** The census (`coupling vs action`) finds coupled bands ($\delta \neq 0$ in the quotient) with *nontrivial* action: 24 of 60 for $D_4 \times C_2$, 72 of 108 for $Q_8 \times C_2$. For these, the governing statement is the master formula ($\star$) alone — a correct on-the-nose identity in $\mathrm{Inn}K$ whose descent to any common coinvariant quotient $M_{\langle\sigma_1,\tau_1,\sigma_2,\tau_2\rangle}$ still carries the twist $(\sigma_1\tau_1)$ on the $\delta(\rho_2)$ term. So:

- **No gap in the posed question.** The question fixes trivial centre-action; there the answer is complete and proved: (B), with $\chi = [A_1, B_2] + [A_2, B_1]$, quadratic law, dichotomy.
- **Deliberately-left edge (not a gap in the theorem):** a *common-receptacle* additivity theorem for bands of differing nontrivial action. The master formula shows the obstruction is the twist $(\sigma_1\tau_1 - 1)\cdot\delta(\rho_2)$ plus the twisted cross terms; packaging this as a genuine cocycle identity on $T^2$ (a cup-product / Baer-sum statement over the full banded gerbe) is the next node, explicitly deferred behind the mini-course per the PROVE brief. It does *not* affect Theorems 1–6.

## 6 Verification summary (hostile-referee pass)

All on `scratch/2026-10-13-delta-additivity.py`, couplers $D_4 \times C_2$ ($|\mathrm{Out}| = 16$), $Q_8 \times C_2$ ($|\mathrm{Out}| = 48$):

| clause | statement | checks ($D_4\times C_2$ / $Q_8\times C_2$) | violations |
|---|---|---|---|
| `[GEN]` | master formula ($\star$) on-the-nose in $M$ | 8113 / 26037 | **0 / 0** |
| central | ($*$) restricted to $\mathrm{Out}^0$ is central | — | True / True |
| `(2)` | $c$ bi-multiplicative on $\mathrm{Out}^0$ | 512 / 512 | **0 / 0** |
| `(2)` over all Out | bi-multiplicativity **fails** off $\mathrm{Out}^0$ | 1408 / 4032 | 112 / 962 *(expected)* |
| `(3)` | $\delta(\rho_1\rho_2) = \delta_1 + \delta_2 + \chi$ on $\mathrm{Out}^0$ | 1905 / 1905 | **0 / 0** |
| `(4)` | $\delta(\rho^n) = n^2\delta(\rho)$ on $\mathrm{Out}^0$ | 320 / 320 | **0 / 0** |
| `(5)` | dichotomy: nonzero & additive | 6 coupled subs each | **0 / 0** |

- **Theorem 1 — GREEN** (hand proof from universal commutator identities + $M$ abelian; computed confirmation 0 violations). The only hypotheses are class $\leq 2$ and common abelian subgroup.
- **Theorems 5–6 — GREEN** (two-line proofs from the central commutator pairing; computed confirmation).
- **Boundary cases.** Trivial band $\rho = 1$: $\chi(1, \rho) = c(1, \bar{B}) + c(\bar{A}, 1) = 0$, additive ✓. $\rho_1 = \rho_2$: $\chi(\rho, \rho) = 2c(\bar{A}, \bar{B}) = 2\delta(\rho)$, consistent with $\delta(\rho^2) = 4\delta(\rho)$ ✓.
- **No RED. No YELLOW** inside the stated scope; the twisted edge of §5 is flagged and is not part of the posed question.

## 7 Grant / program placement

This is the *composition-monoid structure of the waist class*. The butterfly / weak-morphism arc asked whether the weld obstruction composes additively; the answer sharpens the whole gerbe lane:

- **Theory.** The waist class $\delta$ is a *quadratic refinement* on the band composition monoid, with explicit polar form $\chi$ = the commutator pairing of the central extension $\mathrm{Aut}^0 K$. This is the $T^2$ counterpart of the floor $D_1/D_2$ dichotomy and of the two-ends forced correlation: "coupling = nonabelian lift = non-additivity" is a single statement.
- **Applications.** For orchestration / ZS welds, composing two welds does *not* add their obstruction classes; the correction $\chi$ is an economically real interaction term (the "cross-coupling" of two independently-welded subsystems). Additivity holds iff each subsystem is individually uncoupled.
- **Formalisation.** Theorem 1's identity ($\star$) and Lemma 4 are finite, decidable, and a natural Lean target over the existing Inn/Out engine.

**One-line summary.** On $T^2$ the band-to-waist connecting map is a *quadratic map*, not a homomorphism: in general $[A_1A_2, B_1B_2] = [A_1, B_1] + (\sigma_1\tau_1)[A_2, B_2] + \sigma_1[A_2, B_1] + \tau_1[A_1, B_2]$, which on the trivial-action locus reads $\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2) + \big([A_1, B_2] + [A_2, B_1]\big)$ with symmetric bi-additive polar form $\chi$; and $\delta$ is additive on an abelian subgroup **iff** it is identically zero there — a coupled torus is always a witness to non-additivity.